

Lecture 2

Reiterating effort expectations

- Maintain a notebook and take notes. I will be providing prompts for you to do some work; work alongside me;
- It is better to try and get it wrong than ...
- Try to revise the material at the first opportunity after the lecture.
- Learning takes time. It is important to keep give time for material Last minute cramming is neither efficient nor productive.

- Recap Random Variables, Independent Random Variables, Gaussian and Normal Densities, Expectation, Variance, Law of Large Numbers Central Limit Theorem.
- Much of the material is covered in Sections 6.1–6.4 of the text *Introduction to Probability and Statistics for Engineers and Scientists*, by Sheldon Ross.

Estimators: Point and Interval Estimate

- Consider a population. Interest is to estimate a parameter associated with the population.
- First step is to obtain n **random** samples from the population and measure the parameter of interest.
- Let X_i be the random variable denoting the possible value from the i -th sample.
- Let θ be the parameter of interest.
- Let $x = \{x_1, x_2, \dots, x_n\}$ be values from the sample measurements
- x_i is *realisation* of random variable X_i .
- Any function of the samples x is called a *statistic*.

Estimators: Point and Interval Estimate

- $\hat{\theta} = g(x_1, x_2, \dots, x_n)$ is a *point estimate* of θ .
- $(E(\hat{\theta}) - \theta)$ is called the **bias** of the estimate.
- $E\left((\hat{\theta} - \theta)^2\right)$ is called the mean squared error of the estimate.
- An alternative to a point estimate is the *interval estimate* described using $\theta_l, \theta_u, \gamma$

$$\text{Prob}(\theta_l \leq \theta \leq \theta_u) \geq \gamma$$

- γ is called the *confidence level* and (θ_l, θ_u) is the *interval estimate*.

Recap: Sums of Independent Random Variables

- Consider n the sum of n independent and identically distributed (iid) random variables— $X_i, i = 1, \dots, n$.
- Let $E(X_1) = \mu$ and $\text{VAR}(X_1) = \sigma^2$.
- Let the sample sum be $S_n := \sum_{i=1}^n X_i$.
- We know that $E(S_n) = n\mu$ and $\text{VAR}(S_n) = n\sigma^2$.
- From the Law of Large Numbers, $\frac{S_n}{n} \longrightarrow \mu$,
 - Intuitively, if we take a large number of samples, the sample mean will ‘approach the true mean.

Recap: Sums of Independent Random Variables

- From Central Limit Theorem, with Z being the unit Normal,

$$\text{Prob}\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} < y\right) \rightarrow \text{Prob}(Z < y)$$

- As n becomes large, the distribution of the sum approaches that of the Gaussian distribution with same mean and same variance as the sum.
- Thus for large n , the Gaussian is a good approximation for the distribution of the sum of n iid random variables.

The Unit Normal

- Z is the zero-mean Gaussian random variable with unit variance.
- If X is a Gaussian random variable with mean μ and standard deviation σ , then $\frac{X-\mu}{\sigma}$ is unit normal.
- The above operation essentially *shifts and scales* the original Gaussian, it normalises it.
- Some ‘unit normal’ probabilities
 - $\text{Prob}(|Z| < 1) \approx 0.68$
 - $\text{Prob}(|Z| < 2) \approx 0.95$
 - $\text{Prob}(|Z| < 3) \approx 0.997$
 - $\text{Prob}(|Z| > 1) \approx 1 - 0.68 = 0.32$

The Unit Normal

- $\text{Prob}(0 \leq Z < 1) \approx ?$
- $\text{Prob}(Z \notin (0, 1)) \approx ?$
- $\text{Prob}(Z \notin (0, 2)) \approx ?$
- $\text{Prob}(Z > z_\alpha) = \alpha$

Percentile	90	95	97.5	99	99.5	99.9	99.95
α	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
z_α	1.28	1.645	1.96	2.33	2.58	3.08	3.27

Table entry for z is the area under the standard normal curve to the left of z .

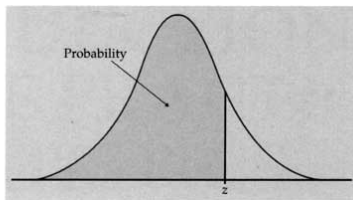


TABLE A Standard normal probabilities (continued)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

Sample Mean

- Assume a population with mean μ and variance σ^2 .
 - μ is unknown and is to be estimated, σ is known.
 - Neither is known and are to be estimated.
- ‘Obvious method’: Obtain n independent samples whose values will be $X_1, \dots, X_n$ and calculate the sample mean as an estimator for the mean.

$$\text{Sample Mean} := \hat{\mu}_n := \frac{\sum_{i=1}^n X_i}{n}$$

- Clearly, $\hat{\mu}_n$ is a random variable. Characterising this random variable characterises the quality of the estimate.

Sample Mean

- The expectation and variance of the sample mean are

$$E(\hat{\mu}_n) = E\left(\frac{\sum_{i=1}^n X_i}{n}\right)$$

Sample Mean

- The expectation and variance of the sample mean are

$$E(\hat{\mu}_n) = E\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{\sum_{i=1}^n E(X_i)}{n}$$

Sample Mean

- The expectation and variance of the sample mean are

$$E(\hat{\mu}_n) = E\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{\sum_{i=1}^n E(X_i)}{n} = \mu$$

Sample Mean

- The expectation and variance of the sample mean are

$$E(\hat{\mu}_n) = E\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{\sum_{i=1}^n E(X_i)}{n} = \mu$$

$$\text{VAR}(\hat{\mu}_n) = \text{VAR}\left(\frac{\sum_{i=1}^n X_i}{n}\right)$$

Sample Mean

- The expectation and variance of the sample mean are

$$E(\hat{\mu}_n) = E\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{\sum_{i=1}^n E(X_i)}{n} = \mu$$

$$\text{VAR}(\hat{\mu}_n) = \text{VAR}\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{\sum_{i=1}^n \text{VAR}(X_i)}{n}$$

Sample Mean

- The expectation and variance of the sample mean are

$$\begin{aligned}E(\hat{\mu}_n) &= E\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{\sum_{i=1}^n E(X_i)}{n} = \mu \\ \text{VAR}(\hat{\mu}_n) &= \text{VAR}\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{\sum_{i=1}^n \text{VAR}(X_i)}{n} = \frac{\sigma^2}{n}\end{aligned}$$

- What is the distribution of $\hat{\mu}_n$?
- How about ‘an approximate distribution’ of $\hat{\mu}_n$?

Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$V = \sum_{i=1}^n (X_i - \mu)^2$$

Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$V = \sum_{i=1}^n (X_i - \mu)^2$$
$$\sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n X_i^2 - n\mu^2$$

Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$V = \sum_{i=1}^n (X_i - \mu)^2$$

$$\sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n X_i^2 - n\mu^2$$

$$E(V) = E\left(\sum_{i=1}^n (X_i - \mu)^2\right)$$

Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$V = \sum_{i=1}^n (X_i - \mu)^2$$

$$\sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n X_i^2 - n\mu^2$$

$$\begin{aligned} E(V) &= E\left(\sum_{i=1}^n (X_i - \mu)^2\right) \\ &= \sum_{i=1}^n E\left((X_i - \mu)^2\right) \end{aligned}$$

Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned}V &= \sum_{i=1}^n (X_i - \mu)^2 \\ \sum_{i=1}^n (X_i - \mu)^2 &= \sum_{i=1}^n X_i^2 - n\mu^2 \\ E(V) &= E\left(\sum_{i=1}^n (X_i - \mu)^2\right) \\ &= \sum_{i=1}^n E\left((X_i - \mu)^2\right) \\ &= n\sigma^2\end{aligned}$$

- Thus if μ is known then V/n is an unbiased estimator of $\text{VAR}(X_1)$.

Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$V = \sum_{i=1}^n (X_i - \hat{\mu})^2$$

Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned}V &= \sum_{i=1}^n (X_i - \hat{\mu})^2 \\ \sum_{i=1}^n (X_i - \hat{\mu})^2 &= \sum_{i=1}^n X_i^2 - n(\hat{\mu})^2\end{aligned}$$

Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$V = \sum_{i=1}^n (X_i - \hat{\mu})^2$$

$$\sum_{i=1}^n (X_i - \hat{\mu})^2 = \sum_{i=1}^n X_i^2 - n(\hat{\mu})^2$$

$$E(V) = E\left(\sum_{i=1}^n X_i^2 - n(\hat{\mu})^2\right)$$

Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$V = \sum_{i=1}^n (X_i - \hat{\mu})^2$$

$$\sum_{i=1}^n (X_i - \hat{\mu})^2 = \sum_{i=1}^n X_i^2 - n(\hat{\mu})^2$$

$$\begin{aligned} E(V) &= E\left(\sum_{i=1}^n X_i^2 - n(\hat{\mu})^2\right) \\ &= \sum_{i=1}^n E(X_i^2) - nE((\hat{\mu})^2) \end{aligned}$$

Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned}V &= \sum_{i=1}^n (X_i - \hat{\mu})^2 \\ \sum_{i=1}^n (X_i - \hat{\mu})^2 &= \sum_{i=1}^n X_i^2 - n(\hat{\mu})^2 \\ E(V) &= E\left(\sum_{i=1}^n X_i^2 - n(\hat{\mu})^2\right) \\ &= \sum_{i=1}^n E(X_i^2) - nE((\hat{\mu})^2) \\ &= n\text{VAR}(X_1) + n(E(X_1))^2 - n\text{VAR}(\hat{\mu}) - n(E(\hat{\mu}))^2 \\ &= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2\end{aligned}$$

Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned}V &= \sum_{i=1}^n (X_i - \hat{\mu})^2 \\ \sum_{i=1}^n (X_i - \hat{\mu})^2 &= \sum_{i=1}^n X_i^2 - n(\hat{\mu})^2 \\ E(V) &= E\left(\sum_{i=1}^n X_i^2 - n(\hat{\mu})^2\right) \\ &= \sum_{i=1}^n E(X_i^2) - nE((\hat{\mu})^2) \\ &= n\text{VAR}(X_1) + n(E(X_1))^2 - n\text{VAR}(\hat{\mu}) - n(E(\hat{\mu}))^2 \\ &= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2 = (n-1)\sigma^2.\end{aligned}$$

- Thus, when μ is not known, $\frac{V}{n-1}$ is the unbiased estimate of $\text{VAR}(X_1)$.